

Operating Systems (CS2018)

Final Exam

Date: June 6th 2026

Course Instructor(s)

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Total Time (Hrs): 3

Total Marks: 100

Total Questions: 4

Do not write below this line

Attempt all the questions.

1. Section B attempt Q1 part b and Sections A and C attempt Q1 part a.
2. All other questions are same for all sections.
3. If any section attempts question no. 1 part of other section, it will mark zero.
4. No formula sheet is allowed.
5. If you feel any ambiguity in the question, assume and state the condition.
6. Do not ask any question from invigilator.
7. Attach question with the answer sheet.

CLO #: 2 Compare elements of processes and threads.

Q1a: Six batch jobs. P1 through P6, arrive at a computer center at different time. Their (externally determined) priorities are 3, 6, 2, 1, 4 and 5, respectively, with 6 being the highest priority. For each of the following scheduling algorithms, draw the complete Gantt chart and compute the mean process turnaround time. Ignore process switching overhead. [25 Marks]

f.a.c.f.s - A.S

Process	Arrival Time	CPU Burst
P1	0	20
P2	3	5
P3	6	14
P4	20	9
P5	22	6
P6	10	24

3
6✓
2
1
4✓
5✓

- (i) Round robin (Time Quantum = 4 ms).
- (ii) Priority scheduling.
- (iii) First-come, first-served.
- (iv) Shortest job first.

→ 7 Marks
3 → Chart
TA 0.5
Aug-1

Q1b: An operating system uses a 3 level multilevel feedback queue scheduling algorithm. [25 Marks]
Queue details:

Q0-Highest priority: Round Robin scheduling: time quantum=4ms

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Q1-Medium priority: Round Robin scheduling: time quantum=8ms
Q2-lowest priority: FCFS Scheduling

Scheduling Rules:

Demotion Rule: If a process uses its complete time quantum without finishing, it is demoted to the next lower queue

Promotion Rule: If a process waits in Q1 or Q2 for more than 12 ms without getting CPU time, it is promoted back to Q0. A promoted process is inserted at the END of Q0.

Process	Arrival Time	CPU Burst
P1	0	20
P2	3	5
P3	6	14
P4	20	9
P5	22	6
P6	10	24

Draw the complete Gantt chart and clearly show: all queue movements all promotion and Demotions (write P and D), any preemption causes by newly arriving processes. Using Gantt chart to compute: average turnaround time average waiting time.

CLO #: 3 Analyze memory management techniques.

Q2a: For each configuration (a-c), identify how many bits are needed for each of the following:

- i. Virtual address
 - ii. Physical address
 - iii. Virtual page number *→ V.A bits - offset*
 - iv. Physical page number
 - v. Offset *→ page size bits*
- a) 32-bit operating system, 4-KB pages, 1 GB of RAM *→ Physical memory*
b) 32-bit operating system, 16-KB pages, 2 GB of RAM
c) 64-bit operating system, 16-KB pages, 16 GB of RAM

Q2b: Consider a system with 8-bit virtual and physical addresses, and 16 byte pages. A process in this system has 4 logical pages, which are mapped to 3 physical pages in the following manner: logical page 0 maps to physical page 6, 1 maps to 3, 2 maps to 11, and logical page 5 is not mapped to any physical page yet. All the other pages in the virtual address space of the process are marked invalid in the page table. The MMU is given a pointer to this page table for address translation. Further, the MMU has a small TLB cache that stores two entries, for logical pages 0 and 2. For each virtual address shown below, explain what happens when that address is accessed by the CPU. Specifically, you must answer what happens at the TLB (hit or miss?), MMU (which page table entry is accessed?), OS (is there a trap of any kind?), and the physical memory (which physical address is accessed?). You may write the translated physical address in binary format. (Note that it is not implied that the accesses below happen one after the other; you must solve each part of the question independently using the information provided above.)

- a) Virtual address 7

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- b) Virtual address 20
- c) Virtual address 70
- d) Virtual address 80

CLO #: 4 Appraise characteristics of file systems.

Q3a: The beginning of a free-space bitmap looks like this after the disk partition is first formatted: 1000 0000 0000 0000 (the first block is used by the root directory). The system always searches for free blocks starting at the lowest-numbered block, so after writing file A, which uses six blocks, the bitmap looks like this: 1111 1110 0000 0000. Examine the bitmap after each of the following additional actions: [10 Marks]

- (a) File B is written, using five blocks.
- (b) File A is deleted.
- (c) File C is written, using eight blocks.
- (d) File B is deleted.

Q3b: Consider a disk that has 10 data blocks starting from block 14 through 23. Let there be two files on the disk: F1 and F2. The directory structure lists that the first data blocks of F1 and F2 are respectively 22 and 16. Given the FAT table entries as below, find which data blocks are allotted to F1 and F2? [10 Marks]

(14, 18); (15, 17); (16, 23); (17, 21); (18, 20); (19, 15); (20, -1); (21, -1); (22, 19); (23, 14).

In the above notation, (x, y) indicates that the value stored in table entry x points to data block y.

Q3c: Explain shared files with the help of a diagram. [5 Marks]

CLO #: 5 Appraise elements of input output techniques.

Q4a: A typical printed page of text contains 50 lines of 80 characters each. Imagine that a certain printer can print 6 pages per minute and that the time to write a character to the printer's output register is so short it can be ignored. Analyze and does it make sense to run this printer using interrupt-driven I/O if each character printed requires an interrupt that takes 50 μ sec all-in to service? [10 Marks]

$$\frac{50 \times 80 \times 60}{60} = 500$$

Q4b: A DMA controller has five channels. The controller is capable of requesting a 32-bit word every 40 nsec. A response takes equally long. Examine that how fast does the bus have to be to avoid being a bottleneck? [10 Marks]

$$\frac{32}{8} = 4 \quad 80 \text{ nsec} \Rightarrow \frac{4}{80 \text{ nsec}}$$

Q4c: Differentiate between precise interrupt and imprecise interrupt. Explain pros and cons of both types of interrupts. [10 Marks]

PL is saved

All prev. instr stored

No new execution needed

Current state is stable

Current state is not defined

Execution are not fully done